# Kartik Malik

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## Professional Summary

Data engineer at Celebal Technologies building enterprise ETL pipelines on Databricks, PySpark, and Azure Data Factory. Databricks Certified Data Engineer Associate with hands-on experience in Medallion Architecture, CDC, and SCD. Proven track record of delivering production-grade data systems across financial analytics and large-scale offline batch processing.

## Experience

**Celebal Technologies** — Jaipur, Rajasthan
*Data Engineer* — *Feb 2026 – Present*

- Engineered resilient ETL pipelines using Databricks, PySpark, and Azure Data Factory by implementing Medallion Architecture, CDC, and SCD techniques, processing enterprise-scale data with full lineage and transformation traceability.
- Optimized financial data processing workflows during Project War Games (PWG 3.0) by developing KPI-driven brokerage analytics pipelines and consolidating six Databricks notebooks into a single production-ready script, reducing deployment overhead and cutting future iteration time by an estimated 60%.

**Wildlife Institute of India** — Dehradun, Uttarakhand
*AI and Software Developer Intern* — *June 2025 – July 2025*

- Built an offline batch data processing pipeline to ingest, validate, and organize 200GB+ of unstructured image data, enabling deterministic grouping and reproducible outputs while preserving original directory structure and metadata.
- Orchestrated Python processing services through a Node.js API layer, improving CPU utilization and sustaining stable processing rates of 50–100 images per minute without cloud or external storage dependencies, eliminating what previously required manual sorting over multiple days.
- **Website**: Link

## Projects

**Business Management System for Food and Beverages** | *React, Node.js, RabbitMQ, PostgreSQL, Docker* — Link

- Designed a relational data model and transactional API layer using PostgreSQL and Prisma, enabling consistent schema migrations and reliable order, cart, and user data handling, reducing query-related failures.
- Implemented a RabbitMQ-driven order processing pipeline that decouples order intake from fulfillment, tripling throughput and supporting concurrent event-driven notifications under simulated peak load without overloading the API server.

**AI Species Recognition System** | *Python, TensorFlow, OpenCV* — Link

- Built a real-time species identification system achieving 90% classification accuracy across 1,500+ animal images, with sub-200ms inference suitable for field deployment.
- Enhanced model generalization by applying preprocessing and augmentation techniques, reducing overfitting and improving performance on unseen data through hyperparameter tuning and validation.

## Technical Skills

**Languages:** Python, SQL
**Data Engineering:** Databricks, PySpark, Azure Data Factory, Delta Lake, ETL Pipelines, Data Modeling, Medallion Architecture, CDC, SCD
**Cloud & DevOps:** Azure, AWS, Docker, Git, Linux
**Databases:** MySQL, SQL Server, PostgreSQL

## Certifications

**Databricks Certified Data Engineer Associate** — Link

- Validated expertise in building ETL pipelines, data transformation, data modeling, and distributed data processing using Databricks and Apache Spark.

**AWS Academy Graduate – Cloud Architecting** — Link

- Gained expertise in cloud architecture principles, design patterns, and best practices using AWS services.

## Education

**University of Petroleum and Energy Studies** — Dehradun, Uttarakhand
*BTech in Computer Science, Major in Cloud Computing and Virtualisation Technology* — *Aug 2022 – May 2026*

**CGPA:** 7.8

## Extracurricular Activities

**Vultr Hackathon — Healthcare AI Project** Link

- Engineered an AI-driven healthcare chatbot by implementing NLP-based intent parsing and symptom-to-guidance mapping, enabling users with limited medical access to receive structured preliminary health support.
- Built a cloud-deployable prototype by designing lightweight backend services and scalable architecture, demonstrating a practical solution for healthcare accessibility and future system expansion.